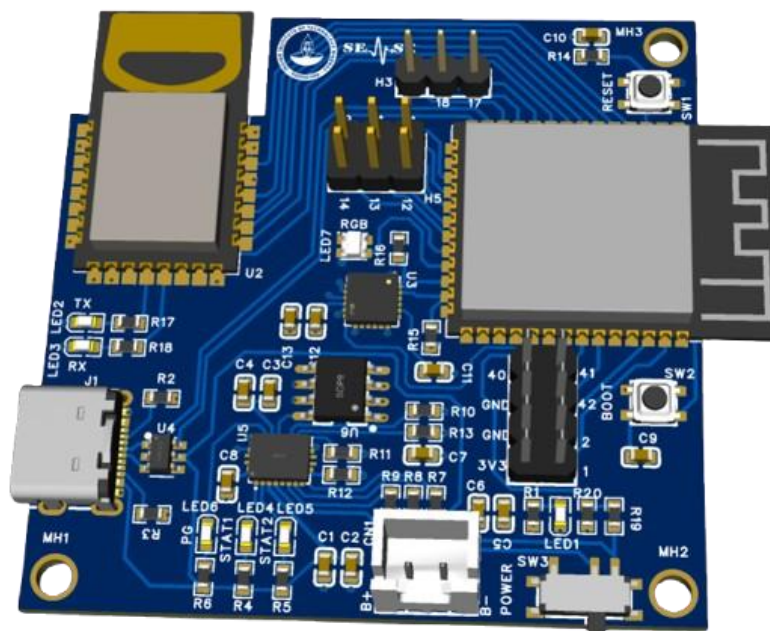


UWB-Based Positioning and Motion Tracking

PCB (V 1.0)



SE-SE

Manufactured by – ARPAN DUTTA

Version -1.0

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❖ Introduction:

This PCB is a compact wireless positioning and motion-sensing development board designed around the ESP32-S3-WROOM-1-N16R8. The board integrates an Ultra-Wideband (UWB) module for precise ranging and localization, an IMU for motion sensing.

The primary objective of the design is to create a portable, battery-powered tracking and sensing platform capable of high-accuracy positioning, orientation estimation, and wireless communication. Such a board can be used for indoor positioning systems, robotics, motion tracking, asset tracking, wearable devices, and IoT applications.

The core processing unit of the system is the ESP32-S3-WROOM-1-N16R8 module, which provides a dual-core processor, Wi-Fi, Bluetooth connectivity, sufficient flash memory, and PSRAM for advanced embedded applications. Positioning and ranging functions are performed by the BU-01 UWB module, while motion and orientation data are acquired through the MPU6050 six-axis IMU sensor.

The board includes a complete power management subsystem consisting of USB Type-C input, battery charging circuitry, voltage regulation, battery monitoring, and power distribution networks, making it suitable for portable and battery-operated applications.

❖ System Description:

1. ESP32-S3-WROOM-1-N16R8 (U1):

The ESP32-S3-WROOM-1-N16R8 serves as the central processing unit (CPU) of the entire system and is responsible for coordinating the operation of all onboard peripherals. Developed by Espressif Systems, the module integrates a high-performance dual-core Xtensa® LX7 processor operating at frequencies up to 240 MHz, making it suitable for real-time embedded applications requiring fast data processing and wireless connectivity.



The module includes 16 MB of external Flash memory and 8 MB of PSRAM, providing sufficient storage for firmware, large application programs, machine learning models, graphics, and sensor data buffering. It also supports secure boot and flash encryption to protect firmware from unauthorized access.

The ESP32-S3 features integrated 2.4 GHz Wi-Fi (IEEE 802.11 b/g/n) and Bluetooth 5 Low Energy (BLE), enabling wireless communication with computers, smartphones, cloud servers, and IoT platforms without requiring additional communication hardware.

The controller provides numerous digital peripherals including GPIO, SPI, I²C, UART, I²S, ADC, PWM, USB-OTG, touch sensing, timers, watchdog timers, and DMA controllers. These interfaces allow seamless communication with external sensors, modules, displays, and storage devices.

2. USB Type-C Interface:

The MUP-U20401-1 - USB Type-C SMT 16 Pin connector serves as the primary interface for powering the PCB, charging the battery, programming the ESP32-S3, and debugging firmware. The reversible connector improves usability and supports modern USB standards.

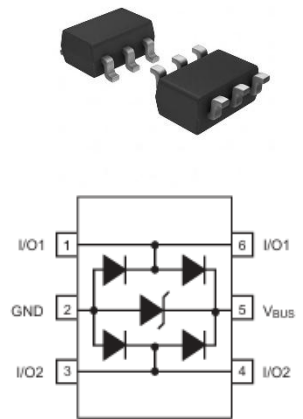


The USB interface is connected directly to the ESP32-S3 USB peripheral, enabling high-speed firmware uploading and serial communication without requiring an external USB-to-UART converter.

3. USBLC6-2SC6 ESD Protection IC:

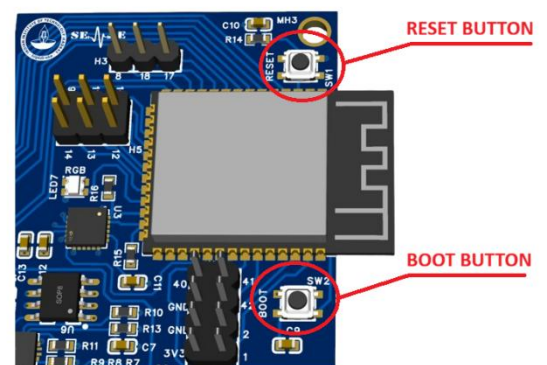
USBLC6 2SC6 is a unidirectional transient voltage suppressor diode designed to protect sensitive USB and low voltage signal lines from electrostatic discharge (ESD) and transient voltage events. It provides fast response and controlled clamping to maintain signal integrity while safeguarding downstream ICs. It has:

- 2 data-line protection.
- Very low capacitance: 3.5 pF max.
- Very low leakage current: 150 nA max.



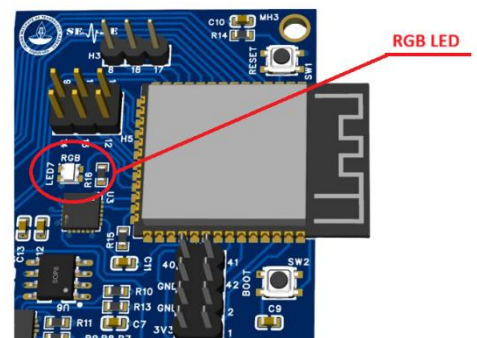
4. Boot and Enable Buttons:

The PCB includes dedicated RESET (SW1) and BOOT (SW2) push buttons for programming and controlling the ESP32-S3. The BOOT button is used to place the microcontroller into bootloader mode, allowing new firmware to be uploaded from a computer. The ENABLE (EN) button acts as a reset button by restarting the ESP32-S3 whenever it is pressed. These buttons simplify firmware uploading, debugging, and system restart during development and testing.



5. WS2812B RGB LED:

The PCB includes a WS2812B-2020 addressable RGB LED connected to **GPIO21** of ESP32-S3 to provide visual indication of the system status. The LED can display multiple colours and lighting patterns under software control by the ESP32-S3. It can be used to indicate power status, successful initialization, communication activity, charging status, errors, and other operating conditions, making debugging and system monitoring easier.



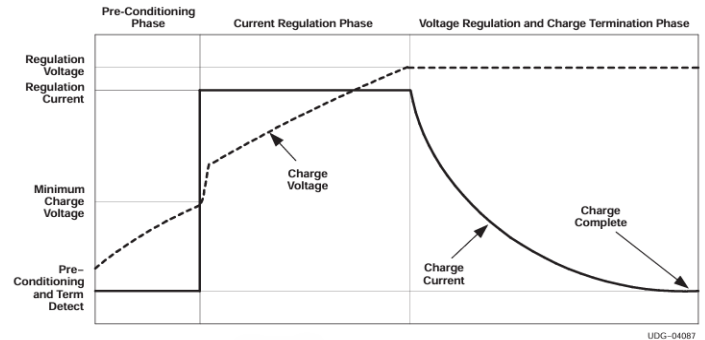
6. BQ24070 Battery Charging and Power Management IC:

The Texas Instruments BQ24070 is a highly integrated battery charging and power-path management integrated circuit designed for single-cell lithium-ion and lithium-polymer batteries. It enables safe battery charging while simultaneously supplying power to the system.



Unlike conventional battery chargers that disconnect the load during charging, the BQ24070 incorporates dynamic power-path management, allowing the PCB to continue operating normally while charging the battery. This feature is particularly beneficial for portable embedded devices that require uninterrupted operation.

The charging process follows a standard three-stage charging algorithm consisting of pre-charge, constant-current charging, and constant-voltage charging. This charging profile maximizes battery lifespan while ensuring safe operation.



The IC includes several built-in protection mechanisms, including:

- Over-temperature protection
- Current limiting
- Short-circuit protection
- Reverse current protection
- Battery fault detection
- Automatic charge termination

For this PCB the Over-temperature protection is disabled but can be added by simply replacing the resistor R11 with a 103AT-type NTC thermistor

In this PCB, the BQ24070 IC's MODE = HIGH; which means the IC is in **Adapter mode** and in this mode, the charge current is set by the resistor on **ISET1** (up to 1.5 A)

• Battery Preconditioning:

During a charge cycle, if the battery voltage is below the $V_{(LOWV)}$ threshold, the bq24070 applies a precharge current, $I_{O(PRECHG)}$, to the battery. This feature revives deeply discharged cells. The R_{SET} resistor, connected between the ISET1 and VSS pins, determines the precharge rate.

$$I_{O(PRECHG)} = \frac{V_{(PRECHG)} \times K_{(SET)}}{R_{SET}} \quad \text{Where, } V_{(PRECHG)} = 0.25 \text{ V} \\ K_{(SET)} = 425$$

In the PCB, R_{SET} (i.e. R12) = 1.5 K Ω

Therefore, $I_{O(PRECHG)} = 70 \text{ mA}$

• Battery Charge Current:

The bq24070 offers on-chip current regulation with programmable set point. The R_{SET} resistor, connected between the ISET1 and VSS pins, determines the charge level.

$$I_{O(OUT)} = \frac{V_{(SET)} \times K_{(SET)}}{R_{SET}} \quad \text{Where, } V_{(SET)} = 2.5 \text{ V} \\ K_{(SET)} = 425$$

In the PCB, R_{SET} (i.e. R12) = 1.5 K Ω

Therefore, $I_{O(OUT)} = 700 \text{ mA}$

- **Charge Termination:**

The bq24070 monitors the charging current through the ISET1 pin during the constant-voltage charging stage. Charging automatically stops when the current falls below the termination threshold, which is 1/10 of the programmed fast-charge current in mode (MODE = High) or 1/25 of the programmed fast-charge current in USB mode (MODE = Low). The RSET resistor connected between the ISET1 and VSS pins sets both the fast-charging current and the charge termination current, $I_{(TERM)}$.

$$I_{(TERM)} = \frac{V_{(TERM)} \times K_{(SET)}}{R_{SET}} \quad \text{Where, } V_{(TERM)} = 250 \text{ mV (Mode = High)} \\ K_{(SET)} = 425$$

In the PCB, R_{SET} (i.e. R12) = 1.5 K Ω

Therefore, $I_{(TERM)} = 70 \text{ mA}$

- **Charge Timer Operation:**

The bq24070 has a safety timer that checks how long the battery is charging. If the battery does not finish charging within the set time, the charger stops and shows a FAULT signal on the STAT1 and STAT2 pins. The resistor connected between the TMR and VSS pins $R_{(TMR)}$ sets the charging time limit.

$$t_{(CHG)} = K_{(TMR)} \times R_{(TMR)} \quad \text{Where, } K_{(TMR)} = 0.36 \text{ S}/\Omega$$

In the PCB, $R_{(TMR)}$ (i.e. R10) = 60.4 K Ω

Therefore, $t_{(CHG)} \approx 6 \text{ Hr}$

- **Dynamic Power-Path Management (DPPM):**

The **DPPM** feature automatically manages the available input power between the system and the battery. When the input power is limited and the system voltage starts to drop, the bq24070 reduces the battery charging current to ensure that the system continues to receive sufficient power. If the system requires more current than the input source can provide, the battery automatically supplies the additional current. This feature maintains a stable system voltage, improves power efficiency, allows the use of lower-power adapters, and ensures uninterrupted operation of the device. $R_{(DPPM)}$ is the external resistor connected between the DPPM and VSS pins, which sets the DPPM voltage, $V_{(DPPM-REG)}$.

$$V_{(DPPM-REG)} = I_{(DPPM)} \times R_{(DPPM)} \times SF \quad \text{Where, } I_{(DPPM)} = 100 \mu\text{A} \\ SF \text{ (scale factor)} = 1.15$$

In the PCB, $R_{(DPPM)}$ (i.e. R13) = 37.4 K Ω

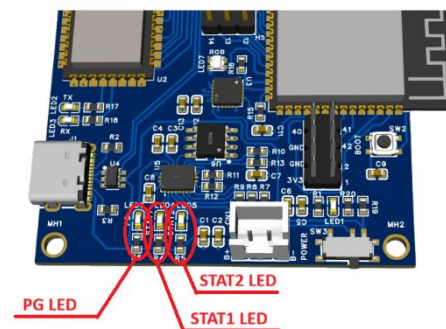
Therefore, $V_{(DPPM-REG)} = 4.3 \text{ V}$

The bq24070 **Power-Path Management** is designed so that when a valid adapter is connected, the system is powered directly from the adapter through the internal power-path transistor (Q1) and the battery is charged independently through another internal transistor (Q2). The battery does not supply current to the system, so it is not being discharged while the adapter is present. This reduces unnecessary charge-discharge cycles and increases battery life. With only the battery connected, OUT can source up to **2 A** continuous safely.

7. Battery Charging Status LEDs:

The PCB includes three status LEDs connected to the PG, STAT1, and STAT2 output pins of the BQ24070 battery charging IC. These LEDs provide a visual indication of the charger's operating status, allowing the user to easily monitor the power and charging conditions.

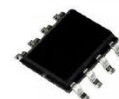
- **PG (Power Good) LED (LED6) (BLUE):** Indicates that a valid input power source, such as a USB adapter, is connected and within the acceptable operating range.
- **STAT1 LED (LED4) (RED):** Indicates the battery charging status. It is typically ON while the battery is being charged.
- **STAT2 LED (LED5) (GREEN):** Works together with the STAT1 LED to indicate different charger states, such as charge complete, charging fault, or standby, depending on the combination of the STAT1 and STAT2 outputs.



CHARGE STATE	STAT1	STAT2
Precharge in progress	ON	ON
Fast charge in progress	ON	OFF
Charge done	OFF	ON
Charge suspend (temperature), timer fault, and sleep mode	OFF	OFF

8. LT1963A Low Dropout Voltage Regulator:

The 4.3V OUTPUT of the charging IC is given to the INPUT of the LT1963A, which is a high-current, low-dropout (LDO) linear voltage regulator that provides a stable 3.3 V power supply to the system electronics. It is capable of supplying output currents up to 1.5 A while maintaining excellent voltage regulation even when the input voltage is only slightly higher than the output voltage.



One of the primary advantages of the LT1963A is its extremely low output noise and excellent load transient response. These characteristics are particularly important for sensitive RF components such as the UWB module and precision sensors like the MPU6050, which require a clean and stable power supply for reliable operation.

- Output Voltage: Fixed at 3.3 V
- Output Current: Up to 1.5 A continuous
- Dropout Voltage: ~340 mV at full load (typical)
- Low Noise: ~40 μ V RMS (10 Hz–100 kHz)
- Quiescent Current: ~1 mA during operation

That means the **minimum input voltage** required to get stable 3.3 V output is:

$$V_{IN(min)} = V_{OUT} + V_{DROP}$$

Typical case: $3.3 \text{ V} + 0.34 \text{ V} \approx 3.64 \text{ V}$

Worst case: $3.3 \text{ V} + 0.5 \text{ V} \approx 3.8 \text{ V}$

9. Battery Voltage monitoring:

The PCB includes a battery voltage monitoring circuit implemented using a resistor voltage divider consisting of R19 (39 K Ω) and R20 (100 K Ω). The divider scales down the battery voltage to a safe level that can be measured by the ESP32-S3 ADC through **GPIO39**. By periodically reading this voltage, the microcontroller can estimate the battery level, monitor battery health, detect low-battery conditions.

- ADC Calculation:

$$V_{ADC} = V_{BAT} \times \frac{100}{39 + 100}$$

Maximum battery voltage

4.2 V

Measured voltage

$$4.2 \times \frac{100}{139} = 3.02V$$

So this voltage is safe for ESP32 ADC.

- The current flowing through the divider at the maximum battery voltage is:

$$I = \frac{V_{BAT}}{R_{Total}}$$

Substituting the values,

$$I = \frac{4.2}{139000} = 30.2 \mu A$$

The total power dissipated by the voltage divider is:

$$P = VI$$

$$P = 4.2 \times 30.2 \mu A$$

$$P = 126.8 \mu W$$

or,

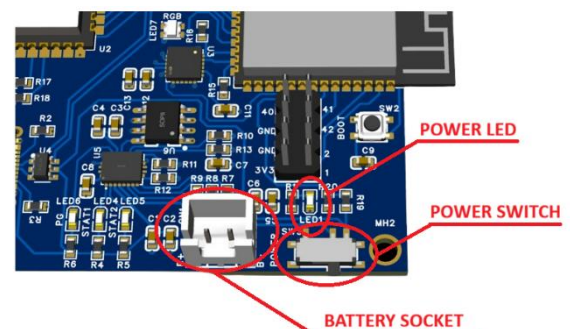
$$P = 0.127 mW$$

10. Power switch, Power LED and Battery socket:

The PCB includes an onboard slide power switch (SW3) that allows the user to manually turn the system ON or OFF. The switch connects or disconnects the battery from the power management circuit, preventing unnecessary battery discharge when the device is not in use.

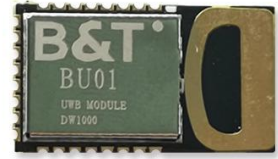
The PCB includes a **Power LED** (LED1) (RED) that provides a visual indication of the system's power status.

Also there is a **2-pin JST XH series connector with a 2.54 mm pitch, straight (vertical) through-hole pins** for connecting a single-cell Lithium-ion (Li-ion) or Lithium-polymer (Li-Po) battery.



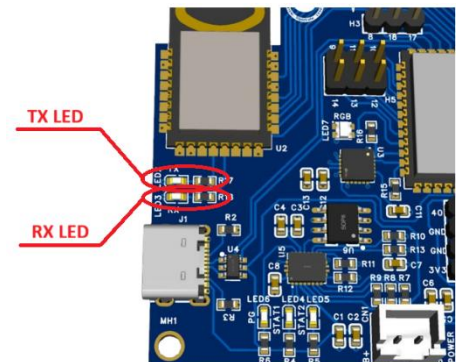
11. BU-01 Ultra-Wideband (UWB) Module:

The **BU-01** is a compact **Ultra-Wideband (UWB)** transceiver module developed by **Ai-Thinker**, designed for high-precision ranging, indoor positioning, and Real-Time Location System (RTLS) applications. It complies with the **IEEE 802.15.4-2011 UWB standard** and operates over the **3.5 GHz to 6.5 GHz** frequency band, enabling centimeter-level distance measurement with excellent resistance to multipath fading and electromagnetic interference. The module integrates the UWB transceiver, RF front-end, and PCB antenna, eliminating the need for external RF design and simplifying hardware development. It operates from a **2.8 V to 3.6 V** supply (3.3 V typical) and supports a **sleep mode current below 1 mA**, making it suitable for battery-powered embedded systems. It communicates with the host microcontroller through a high-speed **SPI interface** and supports configurable data rates of **110 kbps, 850 kbps, and 6.8 Mbps**. It also features **programmable transmit power**, allowing optimization of communication range and power consumption. For localization, the module supports both **Two-Way Ranging (TWR)** and **Time Difference of Arrival (TDoA)** techniques and can be deployed in scalable **RTLS infrastructures** with high tag density. These features make the BU-01 suitable for applications such as asset tracking, robotics, industrial automation, drone navigation, and wearable positioning systems.



12. TX and RX Status LEDs:

This PCB includes dedicated TX (Transmit) status LED (LED2) (RED) and RX (Receive) status LED (LED3) (GREEN) connected to the BU-01 UWB module. These LEDs provide a visual indication of the module's communication activity during wireless ranging and data exchange. The TX LED illuminates whenever the UWB module transmits data packets, while the RX LED illuminates when data packets are received from another UWB device.



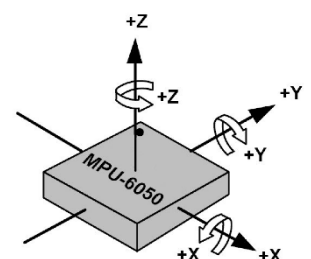
13. MPU6050 Inertial Measurement Unit:

The **MPU6050** is a six-axis MEMS Inertial Measurement Unit (IMU) that integrates a **3-axis gyroscope**, **3-axis accelerometer**, and an embedded **Digital Motion Processor (DMP)** within a single chip. The gyroscope measures angular velocity about the X, Y, and Z axes using MEMS vibratory sensors and digitizes the output through dedicated **16-bit ADCs**, with programmable full-scale ranges of **±250, ±500, ±1000, and ±2000 °/s**. The accelerometer measures linear acceleration along the three axes using capacitive MEMS sensing elements, also with dedicated **16-bit ADCs**, and supports selectable measurement ranges of **±2g, ±4g, ±8g, and ±16g**. The integrated DMP performs onboard motion processing and sensor fusion, reducing the computational load on the ESP32-S3 while providing accurate orientation and motion data with low latency. The MPU6050 communicates with the ESP32-S3 via the **I²C protocol**, operating as a slave device with an address of **0x68** when the **AD0** pin is connected to ground (or **0x69** when AD0 is high). Its high sampling rate, programmable filters, and low-power operation make it well suited for applications such as motion tracking, gesture recognition, robotics, and navigation.



- **Orientation of Axes:**

The diagram on the left, shows the orientation of the axes of sensitivity and the polarity of rotation. Note the pin 1 identifier (•) in the figure.



❖ Connections and Communication Protocols:

1. BU-01 Connections with ESP32-S3:

BU-01 Pin	ESP32 Pin	Function
MOSI	GPIO11	SPI MOSI
MISO	GPIO13	SPI MISO
CLK	GPIO12	SPI Clock
CS	GPIO10	SPI Chip Select
IRQ	GPIO4	Interrupt
RST	GPIO14	Reset
WAKEUP	GPIO15	Wake-up control
TX LED	GPIO16	TX Indicator
RX LED	GPIO17	RX Indicator

• BU-01 SPI protocol:

The **BU-01 UWB module** communicates with the ESP32-S3 using the **Serial Peripheral Interface (SPI)** protocol. In the developed PCB, the **ESP32-S3 functions as the SPI master**, while the **BU-01 operates as the SPI slave**. The ESP32-S3 controls all SPI communication by generating the clock signal, selecting the BU-01 through the Chip Select line, and initiating all data transfers. The BU-01 responds to the master's commands by returning the requested register contents or processing configuration data.

SPI communication begins when the ESP32-S3 pulls the **Chip Select (CS)** line low (**active-low enable**), selecting the BU-01 for communication. Once the slave is selected, the ESP32 generates clock pulses on the **SCLK** line. During each clock cycle, one bit is simultaneously transmitted from the ESP32 to the BU-01 over the **MOSI** line while one bit is transmitted from the BU-01 to the ESP32 over the **MISO** line. This simultaneous bidirectional transfer makes SPI a **full-duplex** communication protocol.

▪ **SPI Operating Mode:**

The BU-01 operates using **SPI Mode 0**, which is defined by:

Clock Polarity (CPOL) = LOW

Clock Phase (CPHA) = LOW

In this mode:

- The clock remains **LOW** when the bus is idle.
- Data is transmitted on one clock edge and sampled on the **rising edge** of the clock.
- This mode ensures reliable synchronization between the ESP32-S3 and the BU-01 and is the default configuration supported by the UWB module.

2. MPU6050 Connections with ESP32-S3:

MPU6050 Pin	ESP32 Pin	Function
SDA	GPIO8	I ² C Data
SCL	GPIO9	I ² C Clock
INT	GPIO18	Motion Interrupt

- **MPU6050 I²C protocol:**

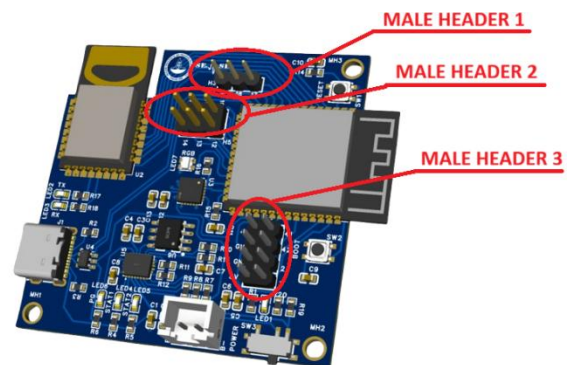
The MPU-6050 supports I2C communications on both its primary (microprocessor) serial interface and its auxiliary interface, but in this case the auxiliary interface is not being used. The **ESP32-S3 acts as the I²C master**, generating the clock signal and controlling all communication, while the **MPU6050 operates as the slave device**.

I²C Address: The MPU6050 supports two selectable 7-bit I²C addresses based on the logic level of the AD0 pin: 0x68 when AD0 is LOW and 0x69 when AD0 is HIGH. In this PCB, the AD0 pin is connected to GND; therefore, the MPU6050 communicates using the I²C address **0x68**.

Since the SDA and SCL lines are **open-drain (open-collector)** outputs, external **pull-up resistors (R15 & R16)** are connected to **3.3 V** to maintain the bus at a logic HIGH level when no device is transmitting.

3. Header Connectors:

The extra GPIO pins of the **ESP32-S3** that are not utilized in the current design are exposed through **three male header connectors** on the PCB to facilitate future expansion and hardware modifications. These headers provide convenient access to unused GPIOs as well as essential power rails, including **3.3 V** and **GND**, enabling additional peripherals and modules to be integrated without redesigning the PCB.



Header	Pins Exposed
MALE HEADER 1 (H3) (1×3)	GPIO8, GPIO18, GPIO17
MALE HEADER 2 (H5) (2×3)	GPIO10, GPIO11, GPIO9, GPIO12, GPIO13, GPIO14
MALE HEADER 3 (H4) (2×4)	GPIO1, GPIO2, GPIO42, GPIO41, 3.3 V, GND (2 pins), GPIO40

The PCB incorporates appropriate **decoupling capacitors** to ensure stable power delivery and minimize supply noise. **10 μF** capacitors are used for bulk decoupling and transient load support, while **0.1 μF (100 nF)** ceramic capacitors are placed close to the power pins of the integrated circuits to filter high-frequency noise and maintain a stable supply voltage. All required resistors and capacitors used throughout the design are implemented in **0603 (1608** in Metric code) surface-mount packages, providing a compact PCB layout while ensuring ease of assembly, good thermal performance, and reliable electrical operation.

❖ Power Consumption:

The PCB operates from a regulated **3.3 V supply** generated by the LT1963A LDO regulator. Therefore, the total power consumption is

$$P_{TOTAL} = V_{SYS} \times I_{TOTAL}$$

Where, $V_{SYS} = 3.3 \text{ V}$ and I_{TOTAL} is the sum of the current drawn by every component.

Current Consumption of Major Components:

Component	Typical Current	Maximum Current
ESP32-S3 (CPU Active)	80 mA	240 mA
BU-01 UWB Module	110 mA	180 mA
MPU6050	3.9 mA	5 mA
LT1963A Quiescent Current	1 mA	1.5 mA
WS2812 RGB LED	20 mA	60 mA (Full White)
Power LED	2 mA	2 mA
Charging Status LEDs (3 LEDs)	0 mA	6 mA
TX/RX LEDs	0 mA	4 mA

- Operating Current:**

During normal operation,

$$\begin{aligned} I_{TOTAL} &= 80 + 110 + 4 + 1 + 20 + 2 \\ &= 217 \text{ mA} \end{aligned}$$

Hence,

$$P_{TOTAL} = 3.3 \times 0.217 = 0.716 \text{ W}$$

Typical board power $\approx$ **0.72 W**

- Worst-Case Current:**

$$\begin{aligned} I_{TOTAL} &= 240 + 180 + 5 + 1.5 + 60 + 2 + 6 + 4 \\ &= 498.5 \text{ mA} \end{aligned}$$

Approximately, $I_{TOTAL} \approx 0.50 \text{ A}$

Thus,

$$P_{TOTAL} = 3.3 \times 0.50 = 1.65 \text{ W}$$

Worst-case board power $\approx$ **1.65 W**

- **Maximum Power from the Power Supply**

The LT1963A can supply Output Voltage of **3.3 V** and maximum Output Current of **1.5 A**

Therefore,

$$P_{MAX} = 4.95 \text{ W}$$

The regulator is therefore capable of supplying nearly **5 W**, while the PCB consumes only about **1.65 W** in the worst case.

- **Design Margin**

$$\text{Current Margin} = 1.5\text{A} - 0.50\text{A} = 1.0\text{A}$$

Or

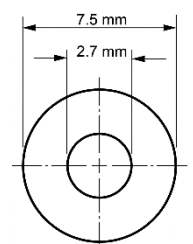
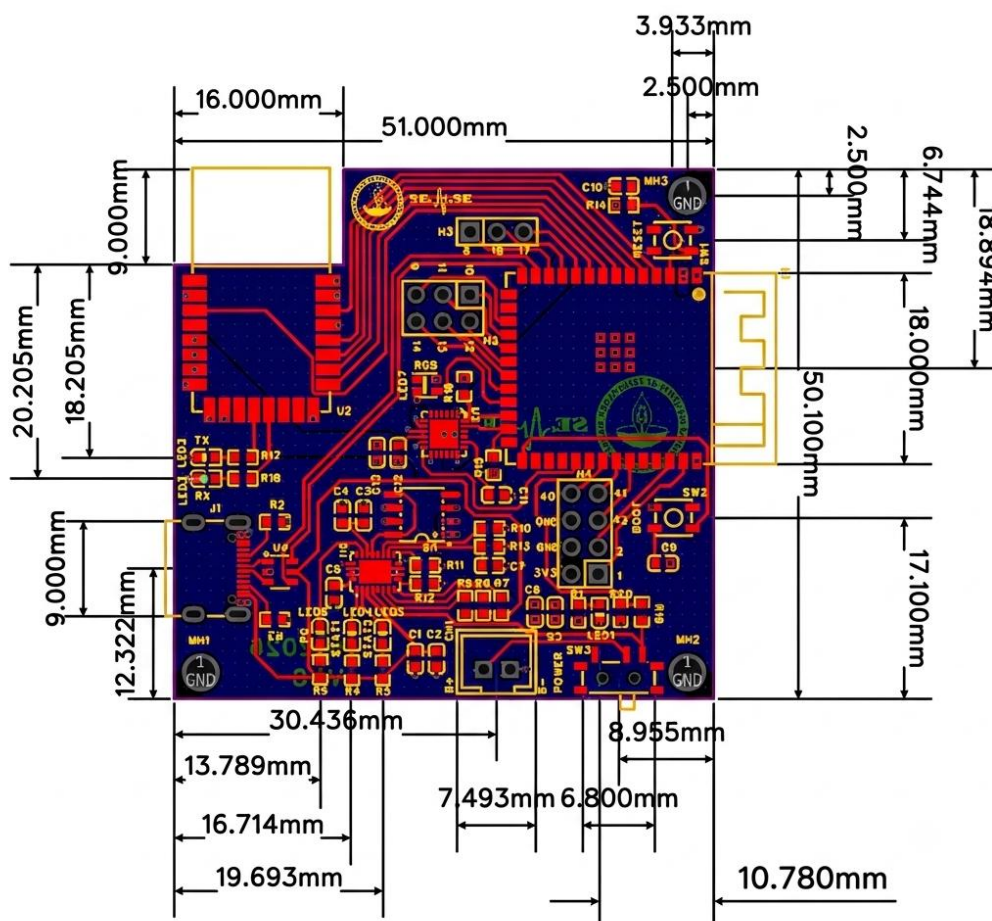
$$\text{Current Utilization} = \frac{0.50}{1.5} \times 100 = 33\%$$

Similarly,

$$\text{Power Utilization} = \frac{1.65}{4.95} \times 100 = 33\%$$

Thus, the regulator operates with approximately **67% headroom**, providing ample margin for transient loads, future peripheral expansion, and ensuring stable operation without approaching its maximum output capability.

❖ Design Dimensions:



- **Mounting Holes:**

The PCB provides **three mounting holes**, each having a **diameter of 2.7 mm**, to enable secure mechanical installation. The hole size is suitable for **M2 screws**, allowing the PCB to be firmly mounted inside an enclosure, onto a chassis, or to other supporting structures

❖ Overall Specifications:

Parameter	Specification
Board Name	UWB-Based Positioning and Motion Tracking PCB
Version	V1.0
Prototype Type	Portable battery-powered UWB positioning and motion tracking development board
Primary Applications	Indoor positioning, RTLS, robotics, motion tracking, wearable devices, asset tracking, IoT
Main Controller	ESP32-S3-WROOM-1-N16R8
Processor	Dual-core Xtensa LX7, up to 240 MHz
Memory	16 MB Flash, 8 MB PSRAM
Wireless Connectivity	Wi-Fi (2.4 GHz), Bluetooth 5 LE, Ultra-Wideband (UWB)
UWB Module	BU-01 (SPI Interface)
UWB Frequency Band	3.5–6.5 GHz
UWB Data Rates	110 kbps, 850 kbps, 6.8 Mbps
Ranging Methods	Two-Way Ranging (TWR), Time Difference of Arrival (TDoA)
Motion Sensor	MPU6050 6-axis IMU
Accelerometer Range	±2g, ±4g, ±8g, ±16g
Gyroscope Range	±250, ±500, ±1000, ±2000 °/s
IMU Interface	I ² C
IMU I ² C Address	0x68 (AD0 = LOW)
Power Input	USB Type-C or Single-cell Li-Ion/Li-Po Battery
USB Input Voltage	4.35 V – 6.4 V (5 V nominal)
Battery Voltage Range	3.0 V – 4.2 V
Battery Type	1S Li-Ion / Li-Po
Battery Charger IC	Texas Instruments BQ24070

Parameter	Specification
Charging Algorithm	Pre-charge → Constant Current → Constant Voltage
Fast Charge Current	700 mA
Pre-charge Current	70 mA
Charge Termination Current	70 mA
Charge Timeout	Approximately 6 Hours
Power Path Management	Dynamic Power Path Management (DPPM)
DPPM Voltage	4.3 V
Voltage Regulator	LT1963A-3.3 V LDO
Regulated Output Voltage	3.3 V $\pm$ 2%
LDO Input Voltage Range	3.64 V – 20 V
Maximum LDO Output Current	1.5 A
LDO Dropout Voltage	~340 mV (typical at full load)
Maximum System Current from Battery	2 A Continuous
Recommended USB Adapter	5 V, 1 A minimum (5 V, 2 A recommended)
Battery Voltage Monitoring	ESP32 ADC via 39 k Ω / 100 k Ω voltage divider
Programming Interface	Native USB Type-C
Debug Interface	Native USB Serial
Communication Protocols	SPI, I ² C, USB, UART, PWM, ADC
UWB SPI Mode	Mode 0 (CPOL = 0, CPHA = 0)
Expansion Headers	3 Male Header Connectors
Status Indicators	Power LED, RGB LED, Charge LEDs (PG, STAT1, STAT2), TX LED, RX LED
Estimated Deep Sleep Current	< 2 mA (board dependent)
Estimated Active Current (ESP32 Only)	80–120 mA
Estimated ESP32 + MPU6050 Current	90–130 mA
Estimated ESP32 + UWB Current	180–260 mA

Parameter	Specification
Estimated Wi-Fi + UWB + MPU6050 Current	320–450 mA
Estimated Absolute Maximum Current Consumption	500–650 mA
Estimated Active Power Consumption	~0.72 W
Estimated UWB Ranging Power	~0.70 W
Estimated Wi-Fi + UWB Power	1.10–1.65 W
Estimated Maximum Power Consumption	≈1.7 W
Operating Supply Voltage (System)	3.3 V
Mounting Holes	3 × M2 (2.7 mm diameter)

❖ Conclusion:

The developed **UWB-Based Positioning and Motion Tracking PCB (V1.0)** integrates the **ESP32-S3, BU-01 UWB module**, and **MPU6050 IMU** into a compact, battery-powered platform capable of high-accuracy indoor positioning, motion tracking, and wireless communication. The inclusion of USB Type-C connectivity, onboard battery charging, voltage monitoring, and expansion headers makes the board suitable for both development and real-world embedded applications.

The power management subsystem, comprising the **BQ24070** battery charger with Dynamic Power-Path Management and the **LT1963A** 3.3 V LDO regulator, provides reliable operation from either USB or a single-cell Li-ion battery. With a typical power consumption of approximately **0.72 W** and a worst-case power requirement of about **1.65 W**, the design operates well within the regulator's **4.95 W** capability, ensuring stable performance and sufficient margin for future expansion.

Overall, the PCB provides a reliable, energy-efficient, and scalable hardware platform for applications including indoor positioning, robotics, asset tracking, wearable devices, and IoT systems.

❖ Relevant Document links:

ESP32-S3-WROOM-1-N16R8: https://www.espressif.com/sites/default/files/documentation/esp32-s3-wroom-1_wroom-1u_datasheet_en.pdf

BU-01 UWB Module: [bu01_product_specification_en_new_v1.0.pdf](#)

MPU6050 IC: <https://invensense.tdk.com/wp-content/uploads/2015/02/MPU-6000-Datasheet1.pdf>

BQ24070 IC: <https://www.ti.com/lit/ds/symlink/bq24070.pdf>

LT1963A-3.3V LDO: [LT1963A Series \(Rev. G\)](#)

USBLC6-25C6: <https://www.st.com/resource/en/datasheet/usblc6-2.pdf>

WS2812B-2020 RGB LED: [WS2812B-2020 Datasheet\(PDF\) - WORLDSEMI CO., LIMITED](#)